COMSATS University Islamabad Wah Campus



Department of Computer Science

Internship Report (Week No. 3)

Submitted By :

Registration No : FA23-BAI-004

Name : Atif Ali Shah

Program : BSAI

Organization : Pakistan Ordnance Factory (POF) Wah Cantt (Rawalpindi)

Department : CO - 4 (IT)

Section : Database Administration (DBA)

Site Supervisor Name : Sir Babar Hussain **Signature :** _____

Submitted To :

Faculty Supervisor Name : Sir Ikram Ul Haq

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Chapter no 5

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Practical No. 05 : Installation of Oracle SQL Developer in Windows and Connecting it with Oracle Database 12c on linux

5.1 Introduction

Oracle SQL Developer is a free Integrated Development Environment (IDE) developed by Oracle Corporation. It provides a graphical user interface (GUI) for interacting with Oracle databases. SQL Developer allows users to execute SQL queries, create and manage database objects, develop PL/SQL programs, and perform basic database administration tasks.

After successfully installing Oracle Database 12c on Oracle Linux, the next step is to install Oracle SQL Developer on Microsoft Windows and establish a connection with the Oracle Database. This enables database administrators and developers to manage the database through a user-friendly graphical interface instead of relying only on command-line tools.

5.2 Objectives

The objectives of this practical were:

- To download Oracle SQL Developer from the official Oracle website.
- To install Oracle SQL Developer on Microsoft Windows.
- To launch SQL Developer successfully.
- To create a new database connection.
- To connect SQL Developer with Oracle Database 12c running on Oracle Linux.
- To verify the database connection.
- To prepare the environment for SQL development and database administration.

5.3 Software Requirements

Software	Version
Microsoft Windows	Windows 10/11
Oracle SQL Developer	Latest Compatible Version
Oracle Database	Oracle Database 12c Release 1 (12.1.0.2)
Oracle Linux	7.9
VMware Workstation	Latest Version

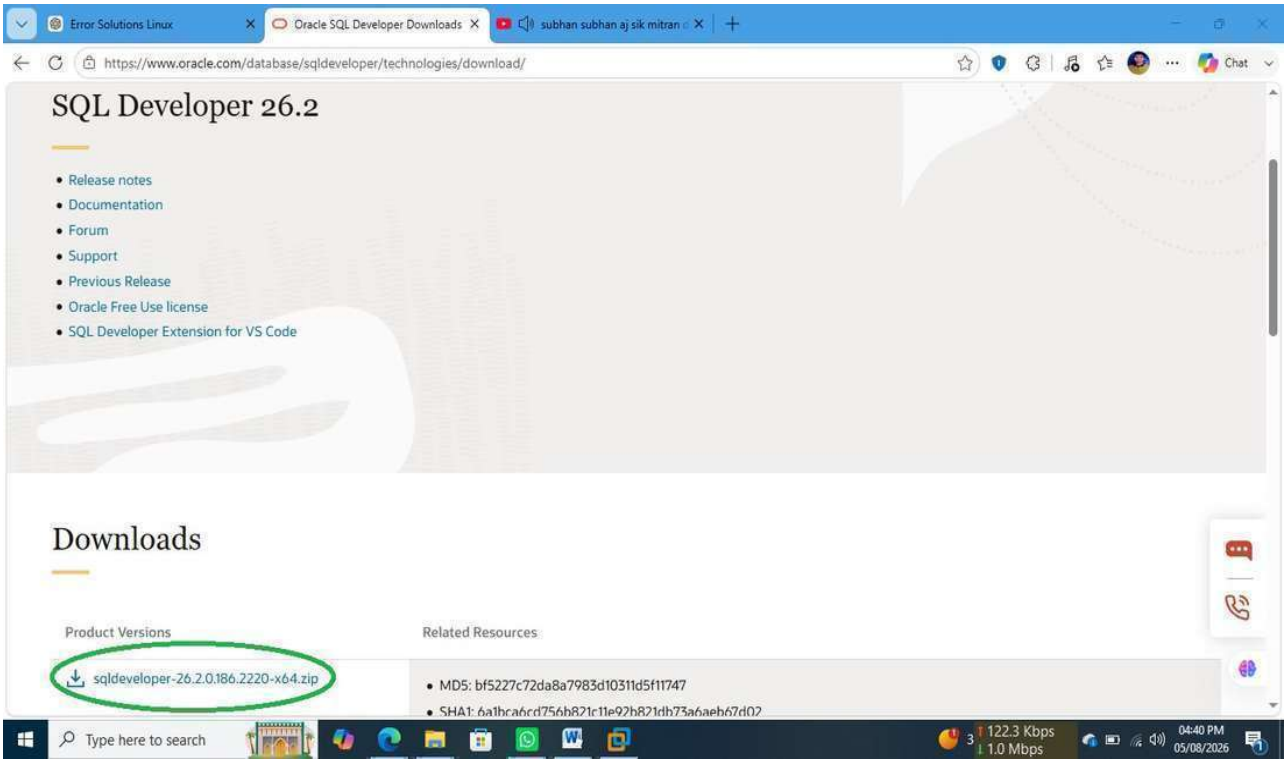
Step 1: Open the Oracle SQL Developer Download Page

Open a web browser on Windows and visit the official Oracle SQL Developer download page.

Explanation

Oracle SQL Developer should always be downloaded from Oracle's official website to ensure that the software is genuine, secure, and compatible with Oracle Database.

Figure 5.3.1: Oracle SQL Developer Download Page



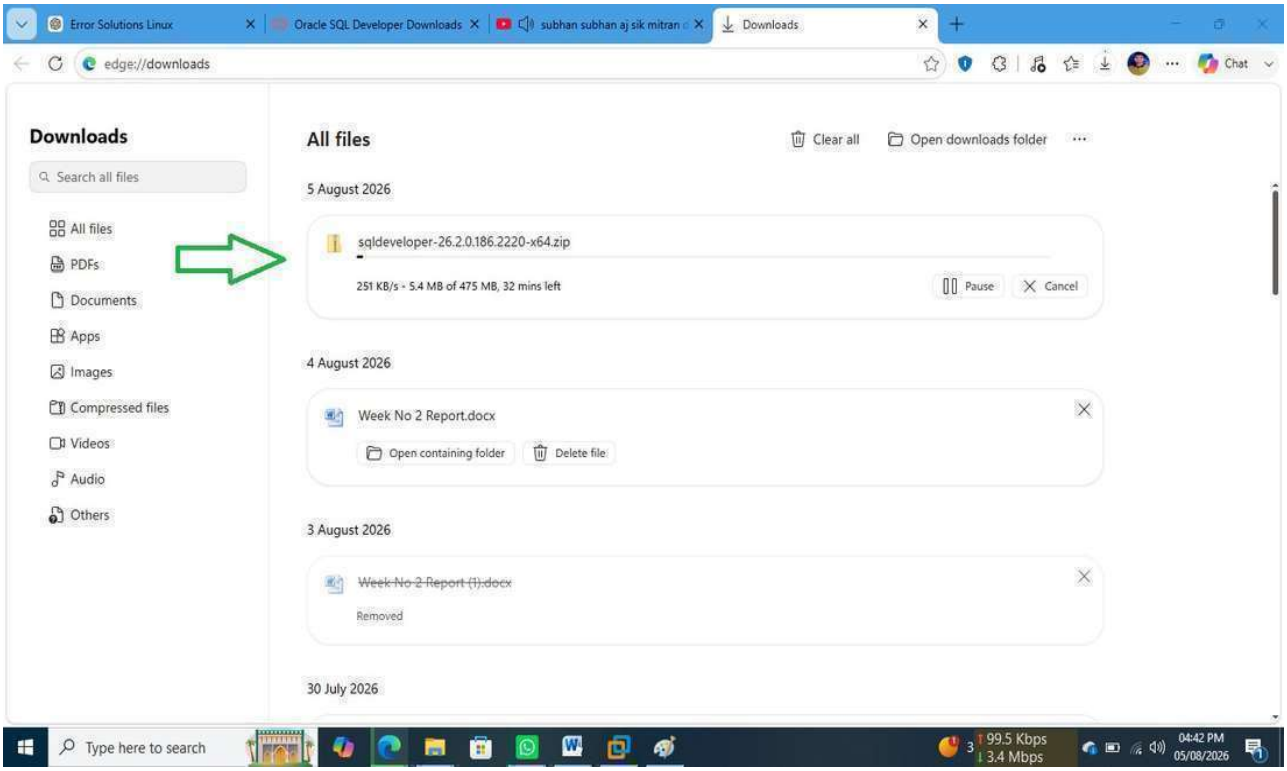
Step 2: Download Oracle SQL Developer

Download the version compatible with your operating system.

Explanation

Choose the appropriate SQL Developer package according to your Windows version. Save the installer in a convenient location such as the **Downloads** folder.

Figure 5.3.2: Downloading Oracle SQL Developer



Step 3: Extract the Downloaded File

If SQL Developer is downloaded as a ZIP archive, extract it.

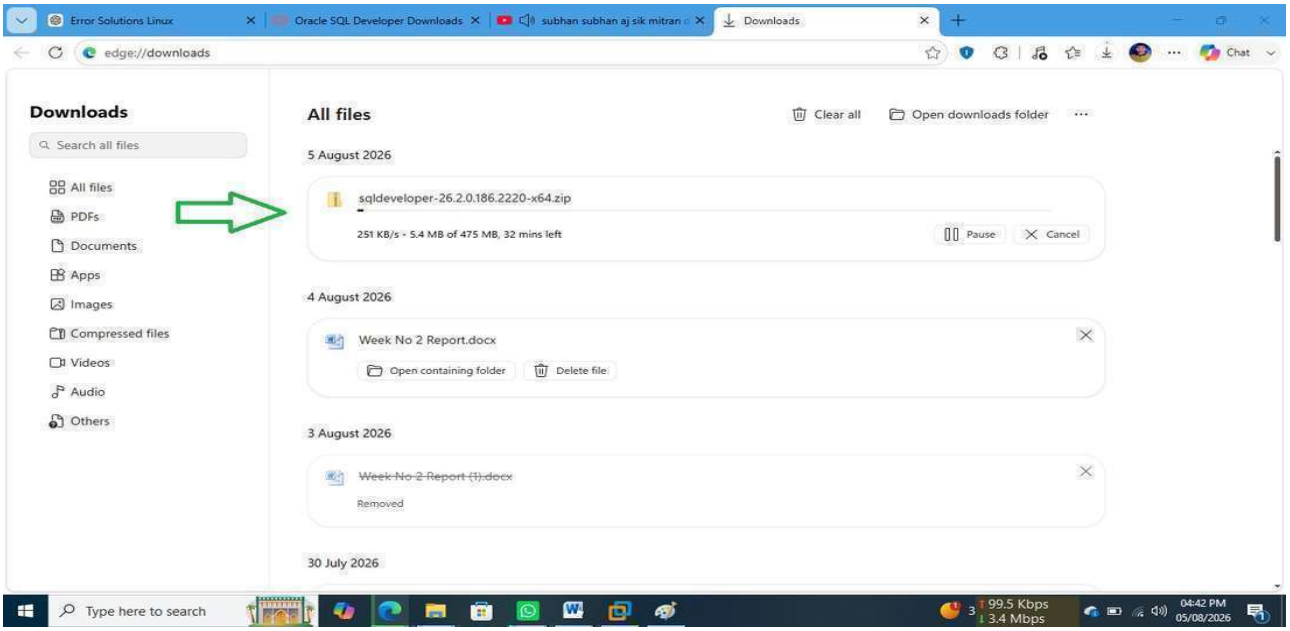
Procedure

- Right-click the downloaded ZIP file.
- Select **Extract All**.
- Choose the destination folder.
- Click **Extract**.

Explanation

Extraction creates the SQL Developer program folder containing all required files.

Figure 5.3.3: Extracting SQL Developer Files



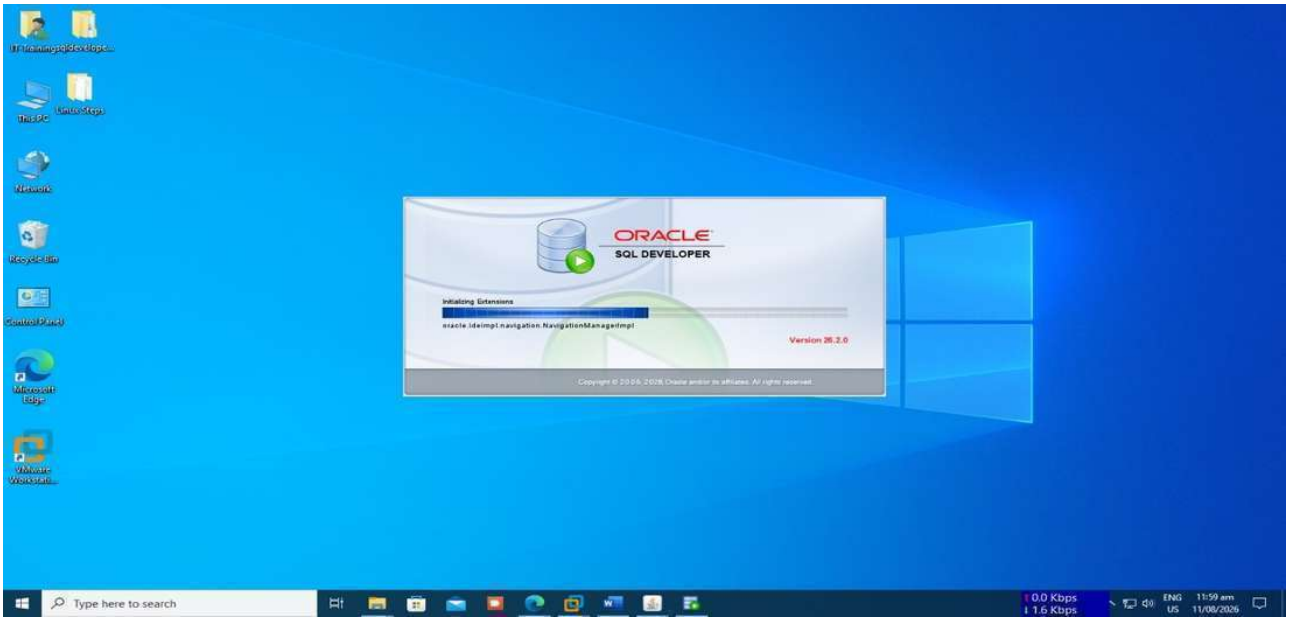
Step 4: Launch Oracle SQL Developer

Open the extracted folder and run **sqldeveloper.exe**.

Explanation

This starts Oracle SQL Developer and loads its graphical user interface.

Figure 5.4.4: Launching Oracle SQL Developer



Step 5: SQL Developer Welcome Screen

Wait for SQL Developer to load completely.

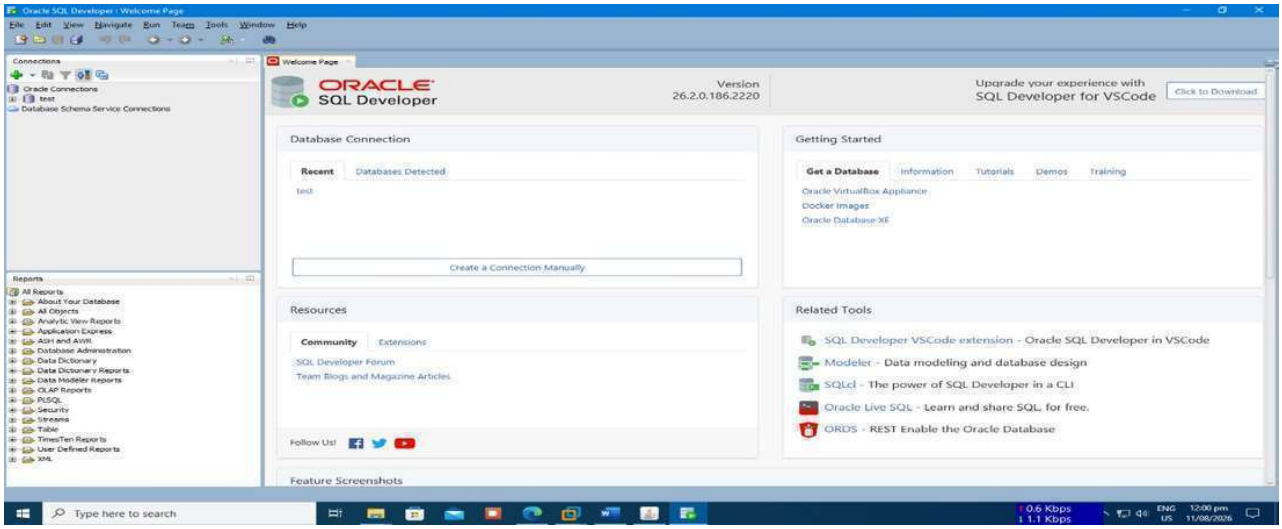
Explanation

The main interface contains:

- Connections Panel
- Worksheet Area
- Menu Bar
- Toolbar
- Reports Panel

This interface will be used throughout the internship to write SQL queries and manage the database.

Figure 5.5.5: Oracle SQL Developer Home Screen



Step 6: Create a New Database Connection

Create a new Oracle database connection.

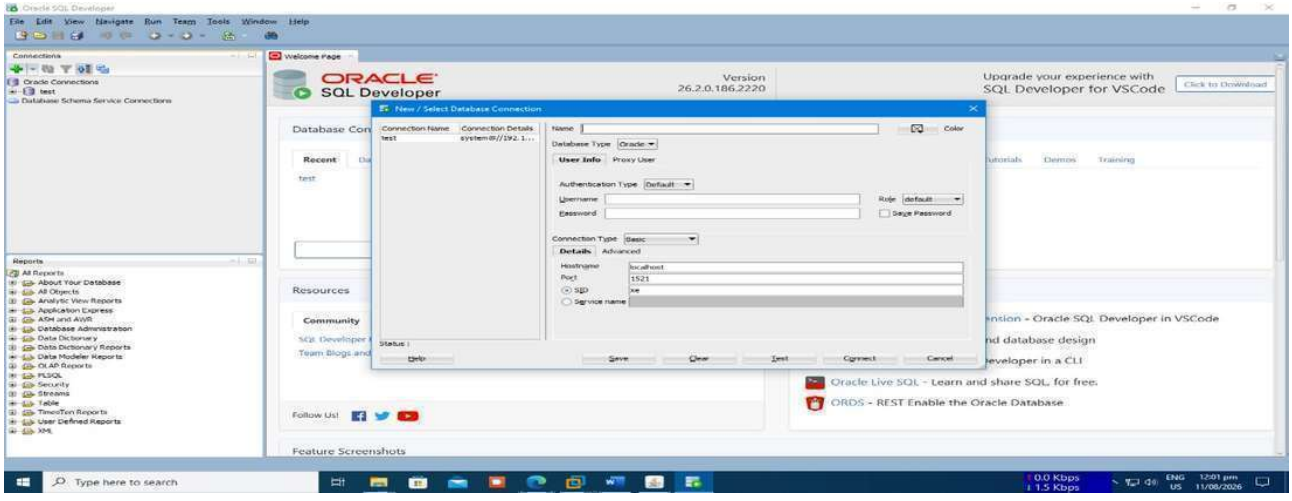
Procedure

- Click the **+ (New Connection)** icon.
- Or go to **File → New → Database Connection**.

Explanation

A database connection stores all the information required to connect SQL Developer with the Oracle Database.

Figure 5.6.6: Creating a New Connection



Step 7: Enter the Connection Details

Fill in the connection window with the appropriate database information.

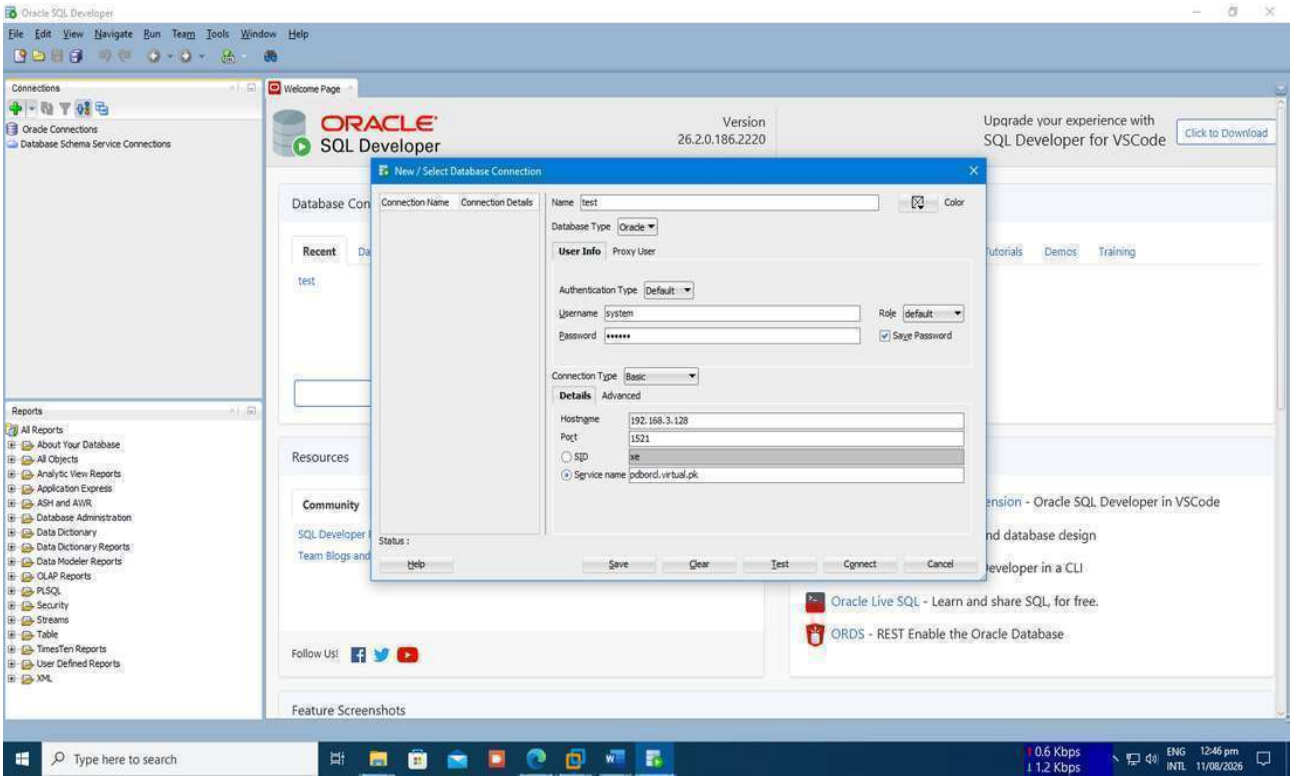
Field	Example Value
Connection Name	Oracle12c
Username	system *(or oracle user provided by your instructor)*
Password	oracle
Save Password	Yes
Connection Type	Basic
Hostname	192.168.x.x *(Replace with your Linux IP address)*
Port	1521
SID / Service Name	orcl *(Use the value configured during installation)*

Note: Replace the hostname, username, password, and SID with the actual values used in your lab environment.

Explanation

These parameters allow SQL Developer to locate and authenticate with the Oracle Database server running on Oracle Linux.

Figure 5.7: Database Connection Configuration



Step 8: Test the Database Connection

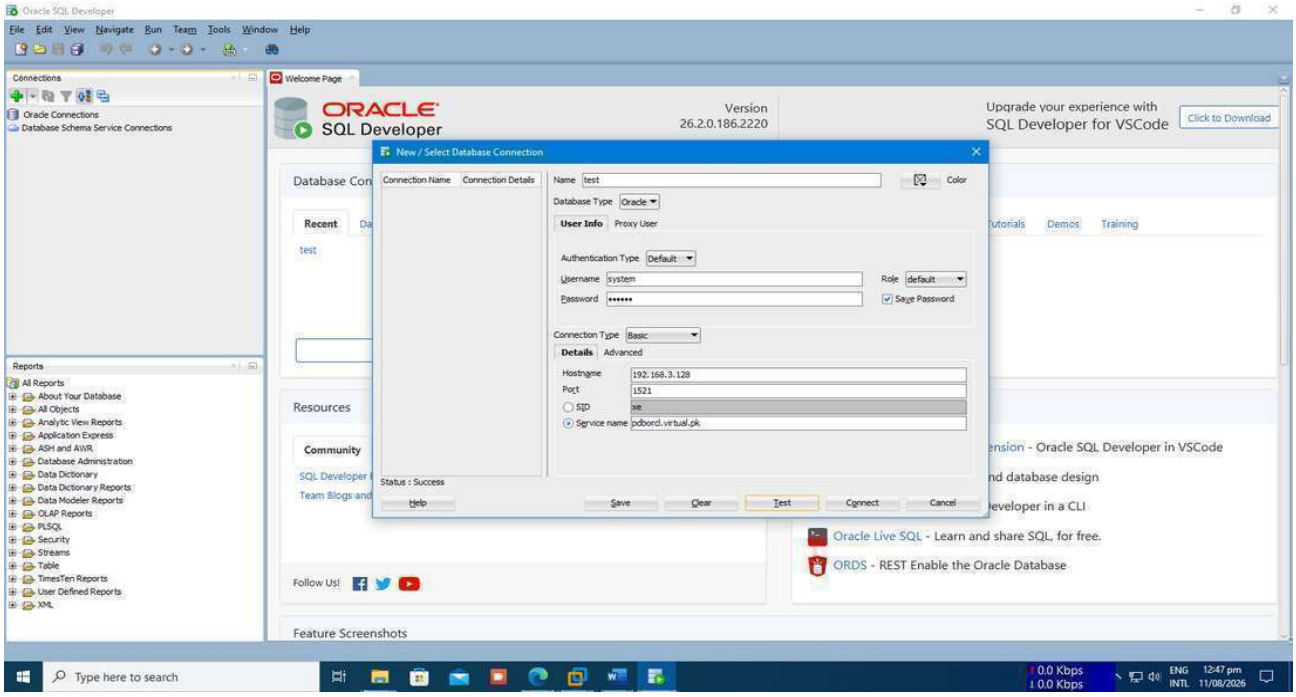
Click the **Test** button in the New Database Connection window.

Explanation

SQL Developer attempts to connect to the Oracle Database using the provided connection details. It verifies the hostname, port number, database service, username, and password.

If all the information is correct, the connection status will be displayed as **Success**.

Figure 5.8: Testing the Database Connection



Step 9 : Verify Successful Connection

If the connection is successful, the status should display:

Status : Success

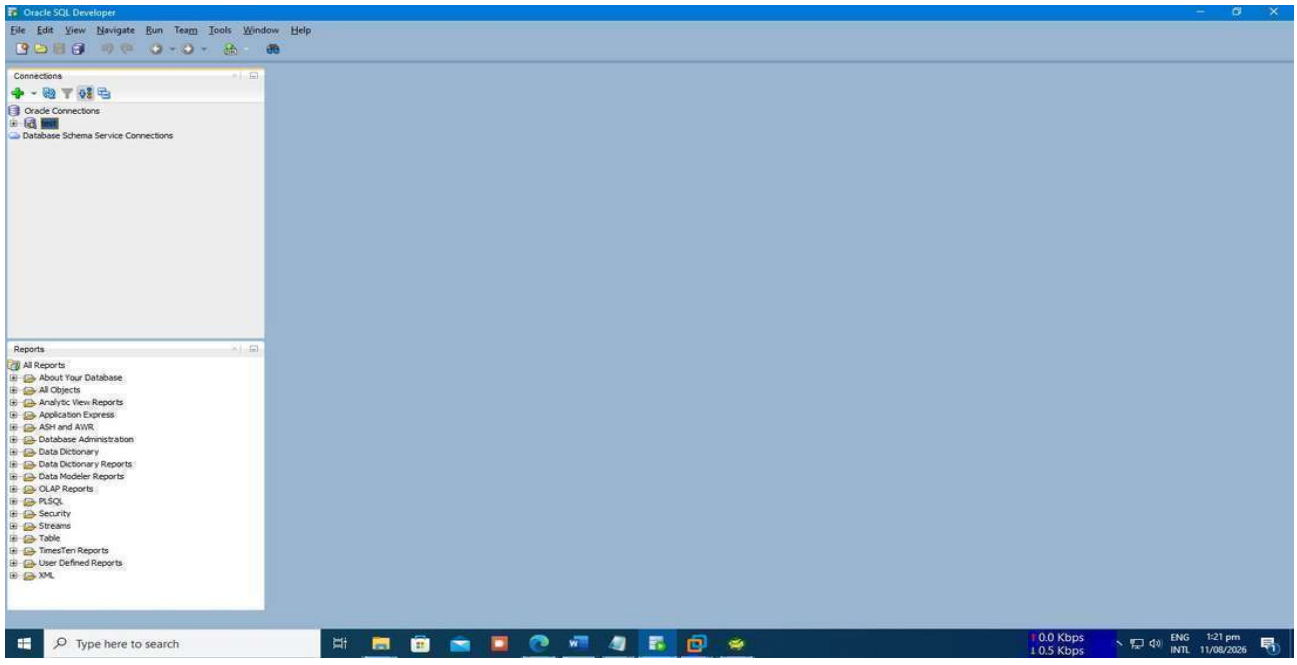
Click the **Connect** button.

Explanation

A successful connection confirms that:

- Oracle Database is running.
- Oracle Net Listener is active.
- The hostname and port number are correct.
- The username and password are valid.
- SQL Developer can communicate with Oracle Database.

Figure 5.9: Successful Database Connection



Step 10: Open the SQL Worksheet

After connecting successfully, SQL Developer opens the database connection.

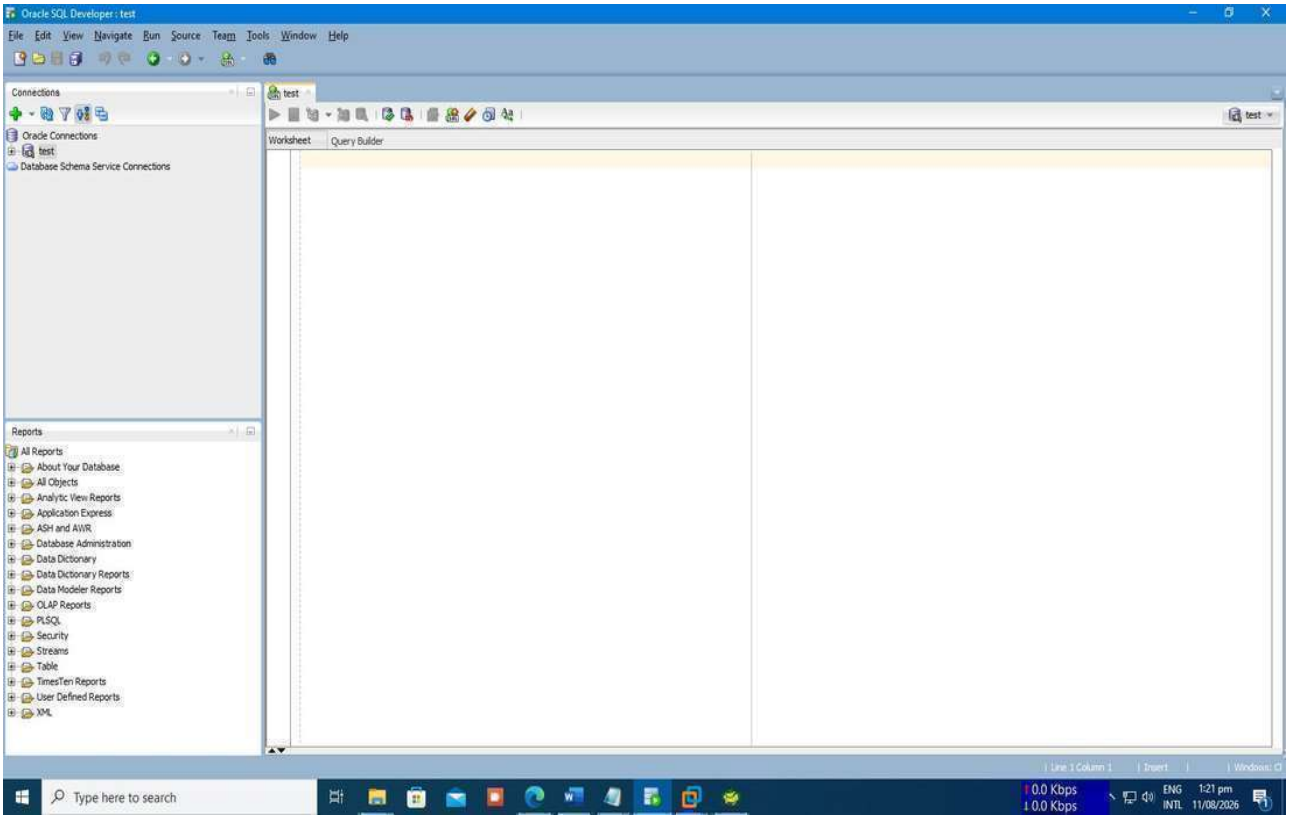
Expand the connection from the **Connections** panel.

Click **SQL Worksheet**.

Explanation

The SQL Worksheet is the working area where SQL statements, PL/SQL programs, and administrative commands are executed.

Figure 5.10: SQL Worksheet



Step 11: Execute the First SQL Query

Execute a simple SQL query to verify that the database is responding correctly.

SQL Command

SELECT * FROM DUAL;

Command Explanation

- SELECT retrieves data from a table.
- DUAL is a special one-row table provided by Oracle for testing SQL statements.

Expected Output

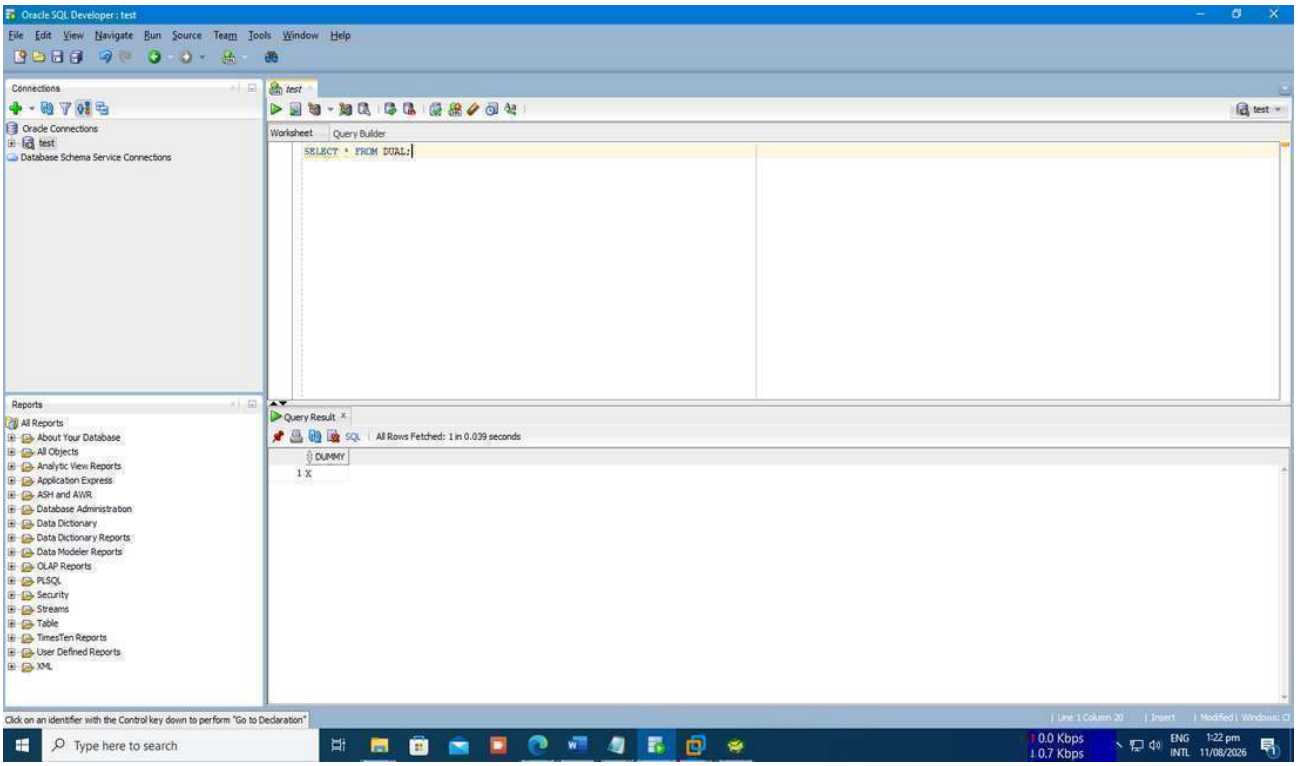
DUMMY

X

Explanation

Receiving the expected output confirms that SQL Developer is successfully connected to Oracle Database and can execute SQL statements.

Figure 5.11: Executing the First SQL Query



Step 12: Display the Current User

Verify the current database user.

SQL Command

SHOW USER;

Expected Output

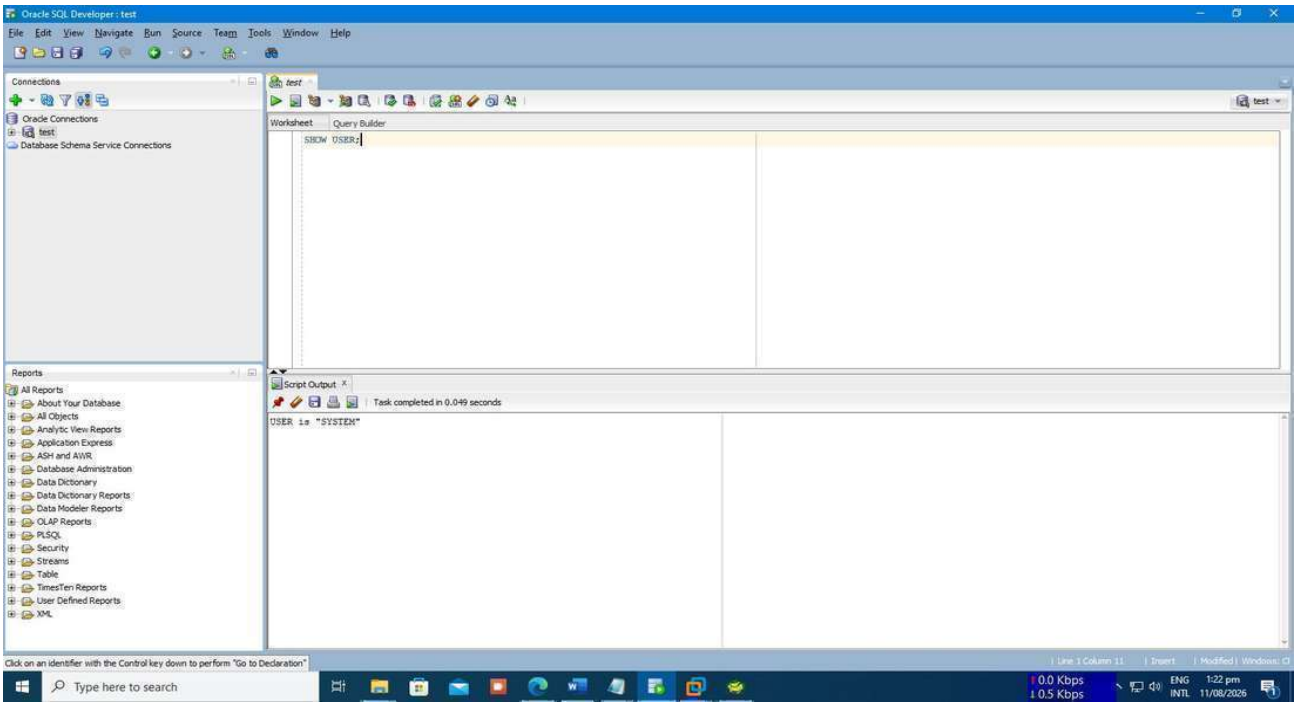
USER is "SYSTEM"

(The username may be SYS, SYSTEM, or another account depending on your connection.)

Explanation

This command displays the username of the current database session.

Figure 5.12: Displaying the Current User



Step 13: Verify the Database Name

Display the name of the connected Oracle Database.

SQL Command

SELECT NAME FROM V$DATABASE;

Expected Output

NAME

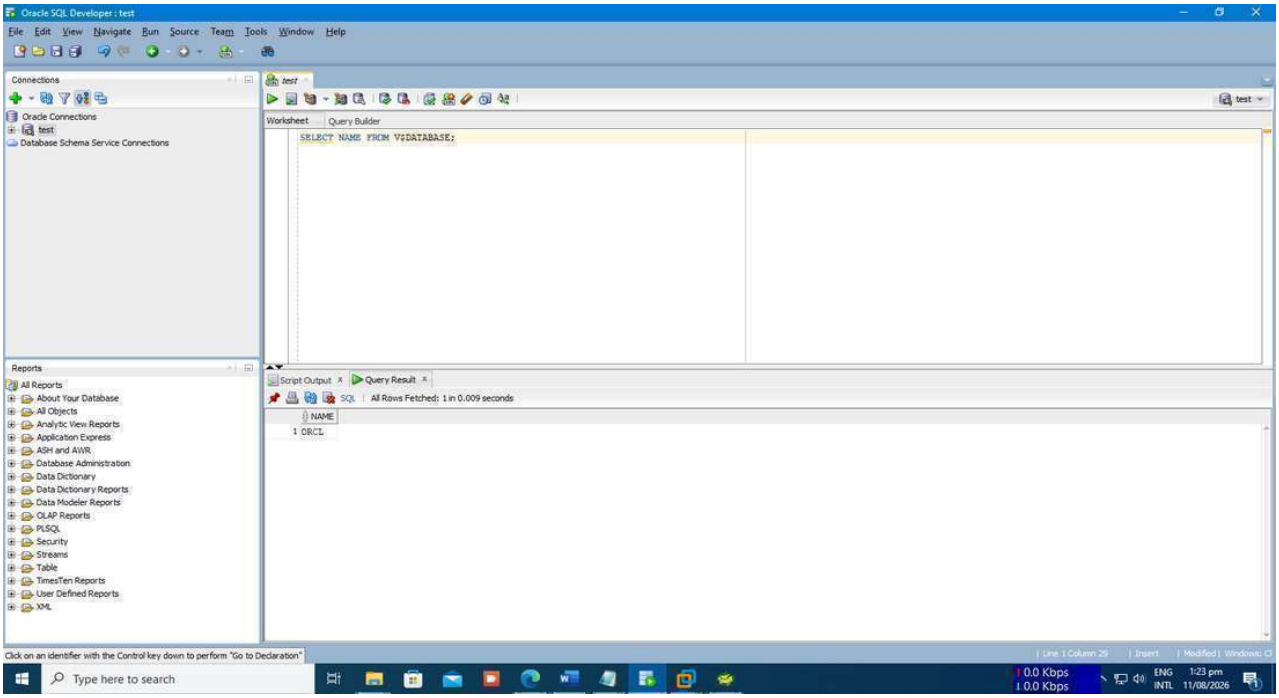
ORCL

(Replace ORCL with your actual database name if it is different.)

Explanation

This command verifies that SQL Developer is connected to the correct Oracle Database instance.

Figure 5.13: Verifying the Database Name



Step 14: Verify the Database Version

Display the Oracle Database version.

SQL Command

SELECT * FROM V$VERSION;

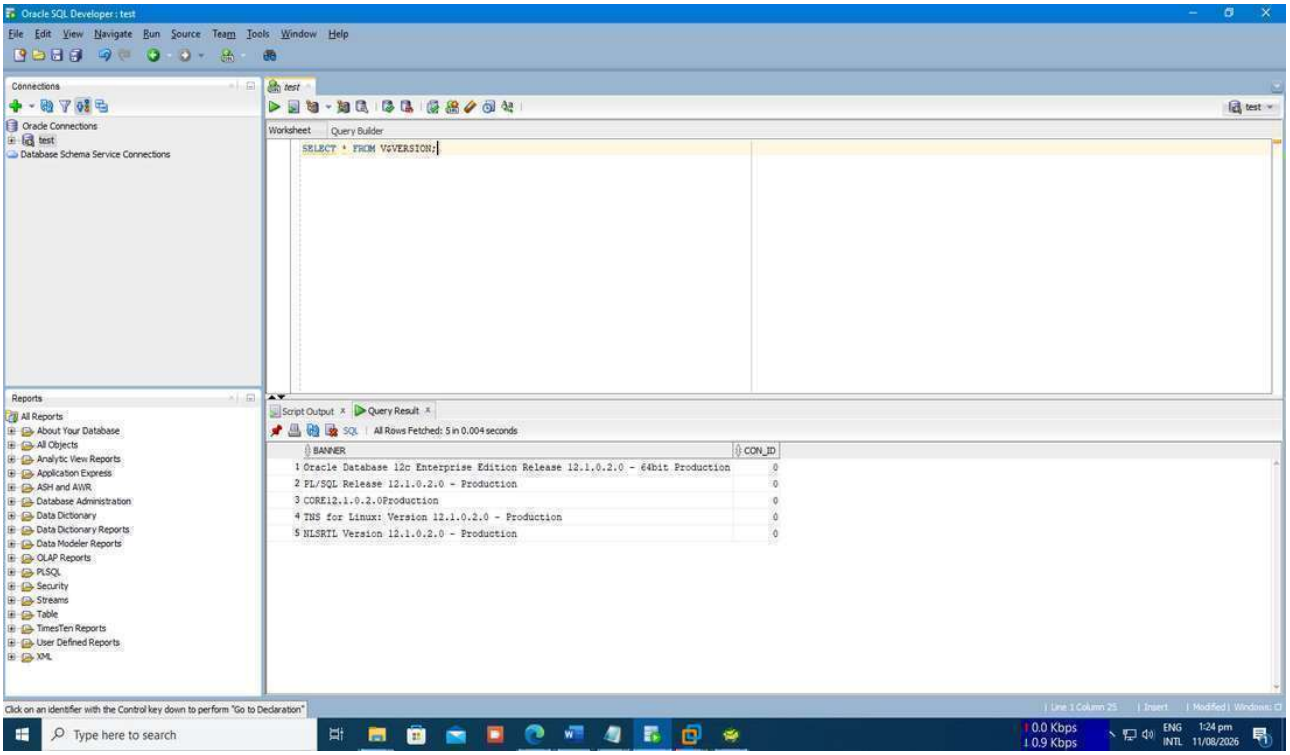
Expected Output

The output will display Oracle Database version information, for example:
Oracle Database 12c Enterprise Edition Release 12.1.0.2.0

Explanation

This command confirms the installed Oracle Database version and verifies that the connection is established with the intended database server.

Figure 5.14: Oracle Database Version



Step 15: Common Connection Errors and Solutions

During the connection process, some common errors may occur.

Error	Possible Cause	Solution
ORA-12541: TNS : no listener	Oracle Listener is not running	Start the Listener using lsnrctl start
ORA-12514	Incorrect SID or Service Name	Verify the SID or Service Name
ORA-01017	Invalid username or password	Re-enter the correct credentials
IO Error: Network Adapter could not establish the connection	Incorrect Hostname/IP Address or Port	Verify the Linux IP address and port number (1521)

Explanation

Understanding these common errors helps a DBA troubleshoot database connectivity issues efficiently.

Key Commands Used in This Practical

SQL Command	Purpose
SELECT * FROM DUAL;	Verify database connectivity
SHOW USER;	Display the current connected user
SELECT NAME FROM V$DATABASE;	Display the database name
SELECT * FROM V$VERSION;	Display Oracle Database version

Practical No. 06 : Toad for Oracle

6.1 Introduction to Toad for Oracle

Toad for Oracle is a database development and administration tool used for working with Oracle databases. It provides a graphical user interface that makes many database development and administration tasks easier to perform.

Toad is developed by **Quest** and is widely used by database developers and database administrators for tasks such as SQL development, schema management, database object management, and administration. Quest provides different editions of Toad with different capabilities.

In this practical, **Toad for Oracle 12.1** is used according to the version recommended by the instructor.

6.2 Role of Toad in Oracle Database Management

Toad can be used by both **Database Developers** and **Database Administrators (DBAs)**.

Its major roles include:

- Executing SQL queries.
- Managing database objects.
- Viewing tables, views, indexes, and other objects.
- Managing users and privileges.
- Monitoring database sessions.
- Viewing schema information.
- Performing database administration tasks.
- Helping developers write and test SQL.
- Providing a graphical interface for Oracle database management.

Therefore, Toad is useful as a graphical database management environment for Oracle.

6.3 Benefits of Toad for Oracle

The major benefits of Toad include:

1. **Graphical User Interface**
   It provides a GUI for performing database tasks.
2. **SQL Development**
   SQL statements can be written, executed, and tested easily.
3. **Database Object Management**
   Database objects can be viewed and managed from the interface.
4. **DBA Support**
   It provides tools useful for database administration.
5. **Schema Management**
   Database schemas and their objects can be explored easily.
6. **Productivity**
   Common database operations can be performed more efficiently through graphical tools.

6.4 Advantages of Toad for Oracle

- User-friendly graphical interface.
- Useful for both developers and DBAs.
- Provides SQL editing and execution features.
- Makes database objects easier to explore.
- Provides administration and monitoring features.
- Reduces the need to perform every task through command-line tools.
- Supports various Oracle database management activities.

6.5 Disadvantages of Toad for Oracle

- It is a commercial software product and licensing may be required.
- Different editions provide different features.
- It can consume more system resources than simple command-line tools.
- Beginners may require some time to learn the interface.
- Older versions may have compatibility limitations with newer environments.
- Some advanced functionality may not be available in every edition.

6.6 Software Requirements

For this practical, the following software is required:

Software	Version/Requirement
Operating System	Windows
Oracle Database	Oracle Database 12c
Toad for Oracle	12.1
Oracle connectivity	Required
SQL Developer	Already installed for database testing

Note: The Toad version used in this practical is **12.1**, as recommended by the instructor.

6.7 Downloading Toad for Oracle 12.1

Step 1: Open the Toad/Quest Website

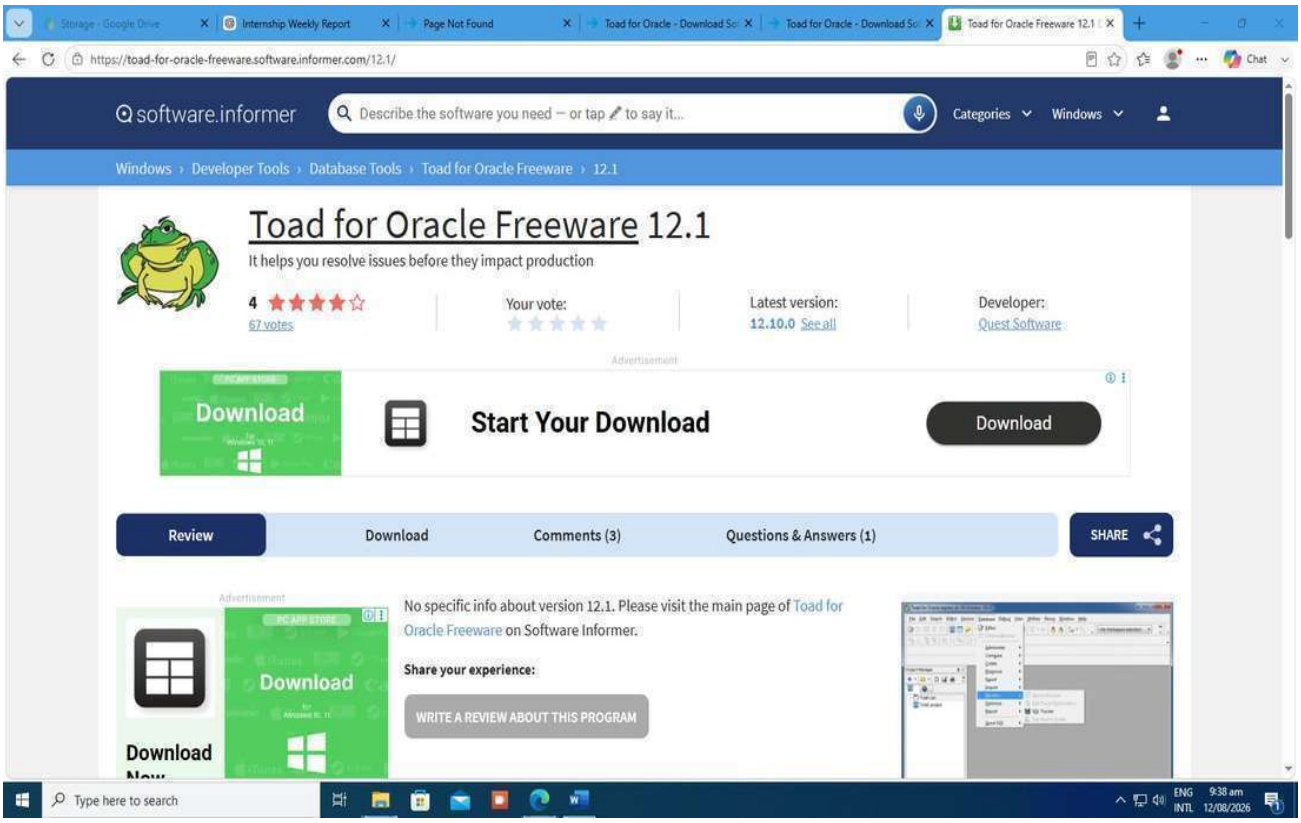
Open a web browser on Windows and navigate to the official Quest Toad for Oracle download/support page.

Quest — Toad for Oracle Downloads

Explanation

Quest is the official vendor of Toad for Oracle. Current Quest download pages require the appropriate entitlement/license access for restricted downloads.

Figure 6.7.1: Toad for Oracle Download Page



6.8 Installing Toad for Oracle 12.1

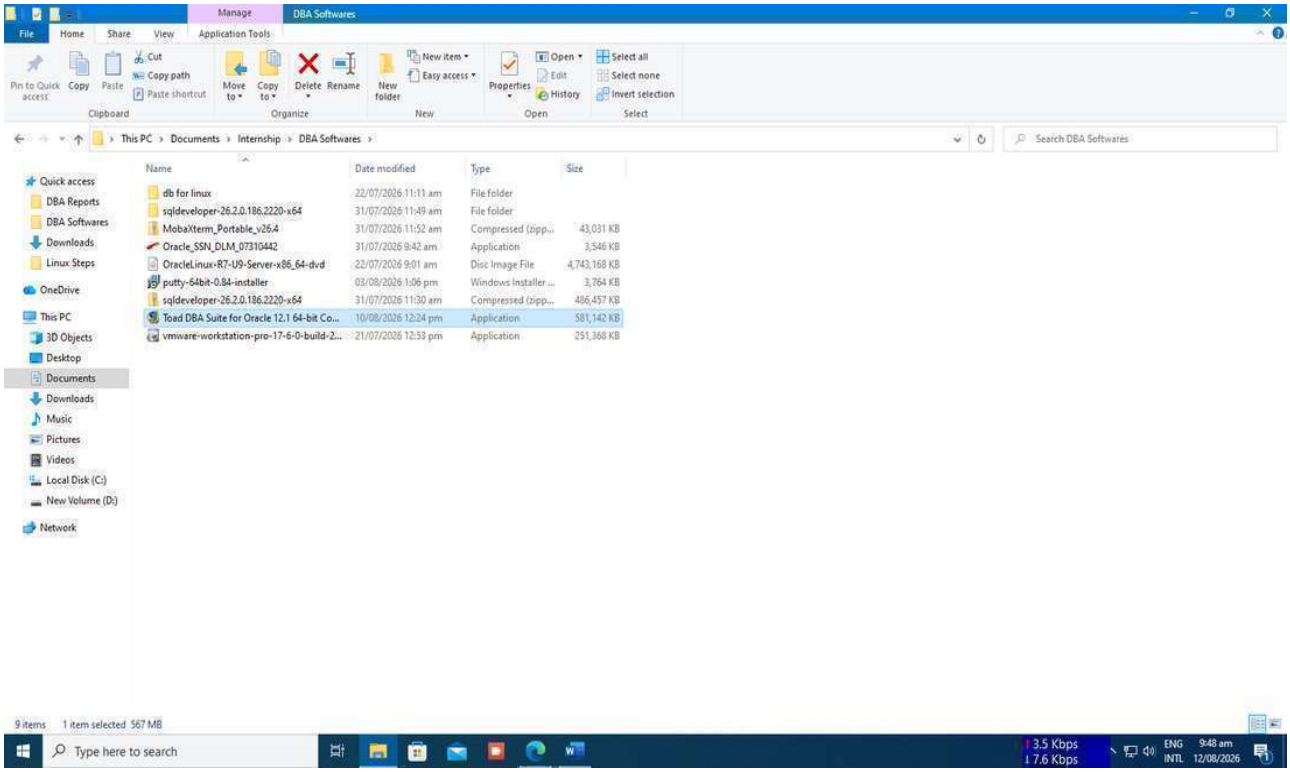
Step 1: Locate the Installer

Open the folder where the Toad installer was downloaded.

Example:

Downloads → Toad for Oracle 12.1

Figure 6.8.1: Toad Installer File



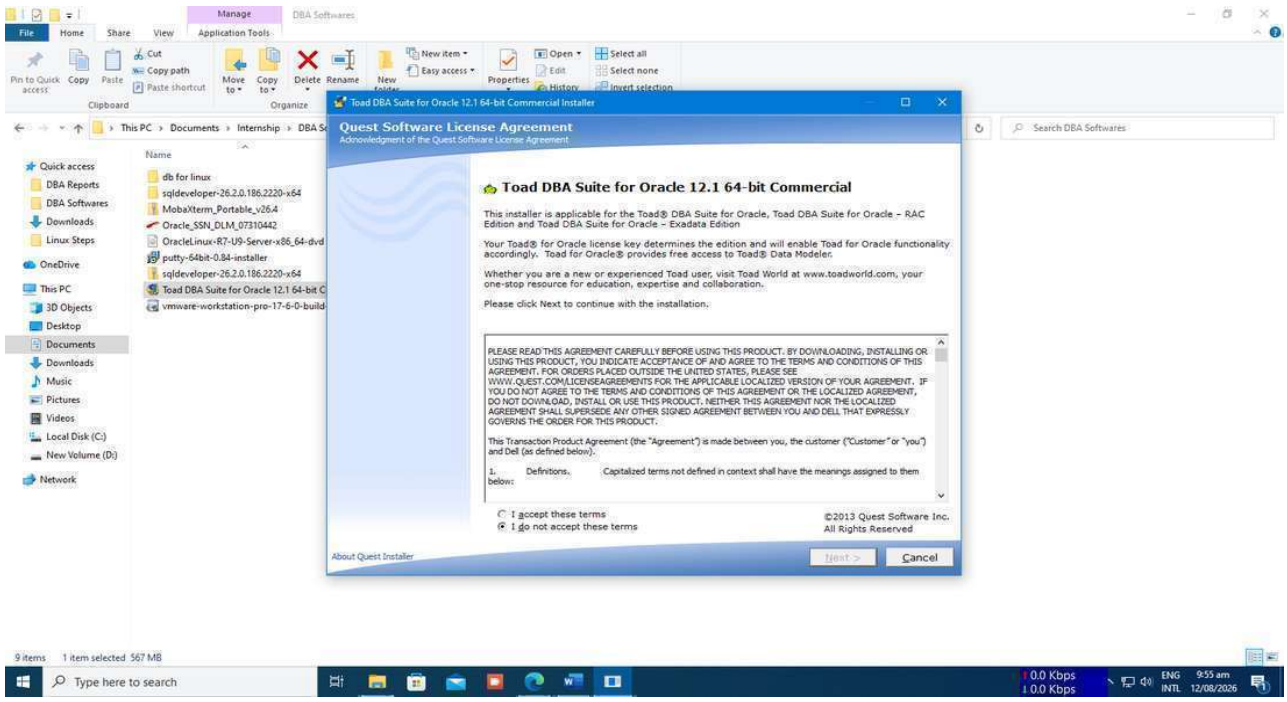
Step 2: Installation Wizard

The Toad installation wizard will appear.

Click:

Next

Figure 6.8.2: Toad Installation Wizard



Step 3: Accept License Agreement

Read the license agreement.

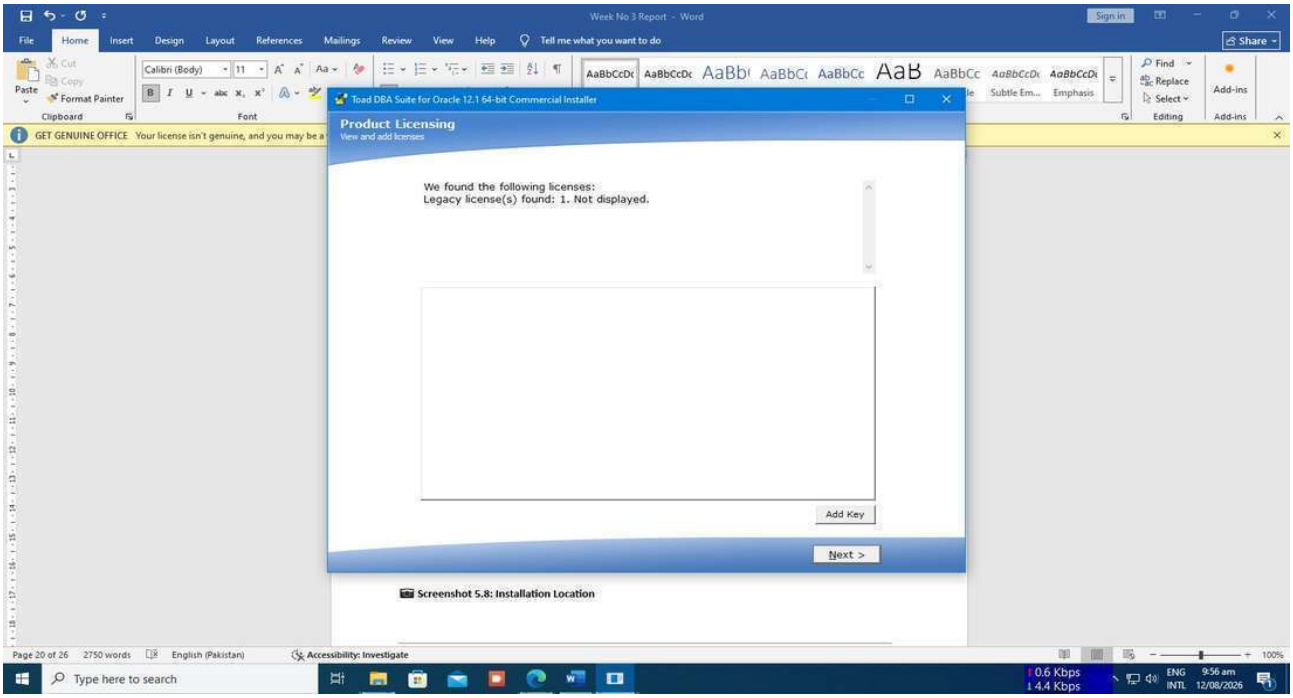
Select:

I accept the terms in the license agreement

Then click:

Next

Figure 6.8.3: License Agreement



Step 4: Select Installation Location

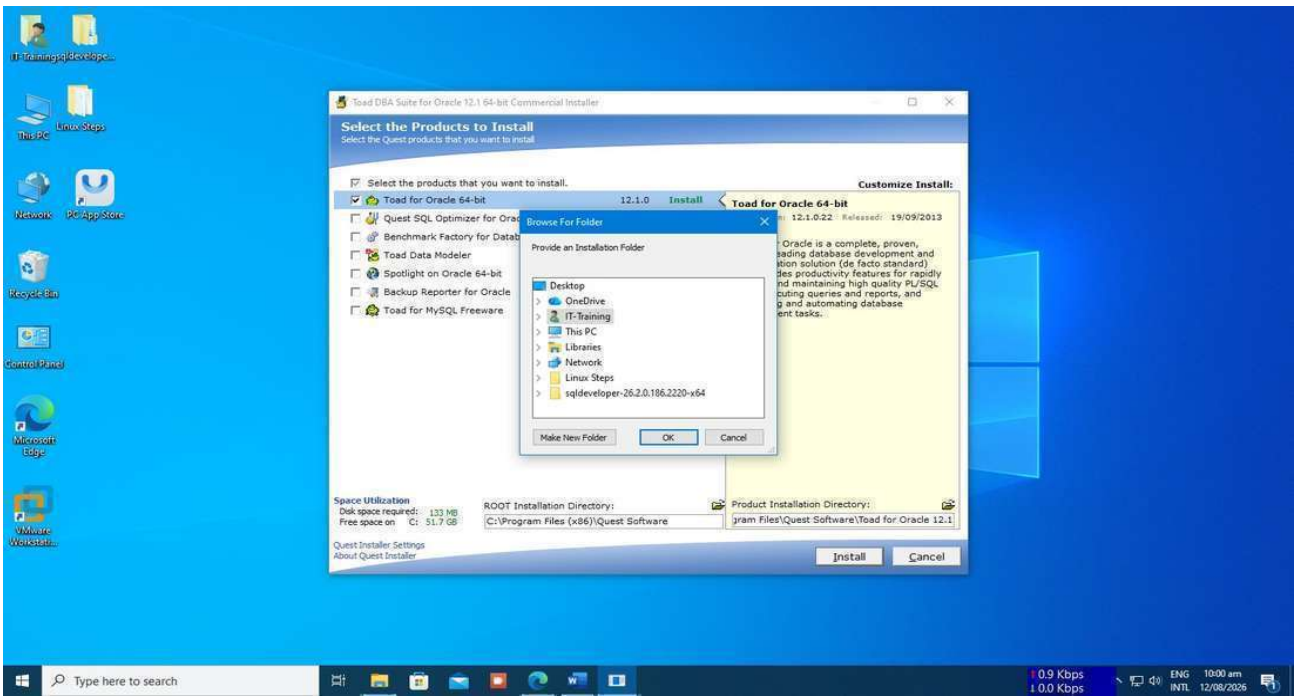
The installer will ask for the installation directory.

Keep the default location unless instructed otherwise.

Then click :

Next

Figure 6.8.4: Installation Location



Step 5: Select Installation Options

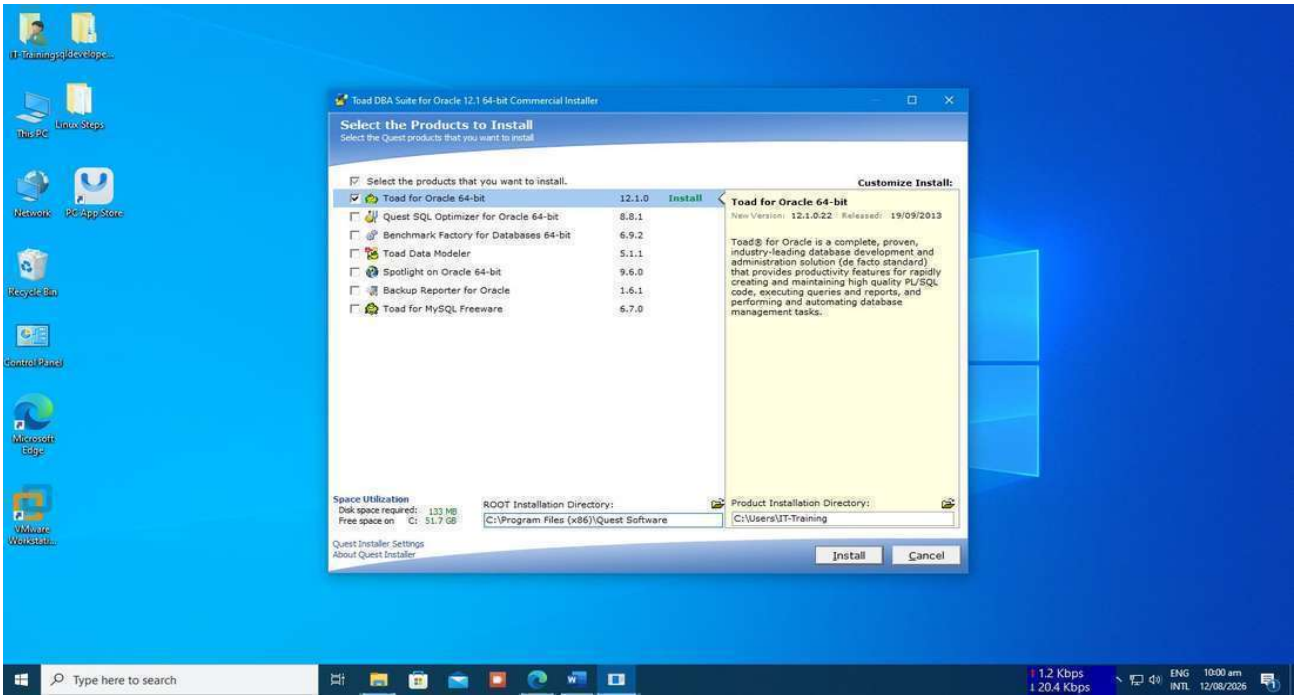
Select the required Toad components/options according to the installer.

For a basic internship setup, the standard/default options can normally be retained.

Click:

Next

Figure 6.8.5: Installation Options



Step 6: Start Installation

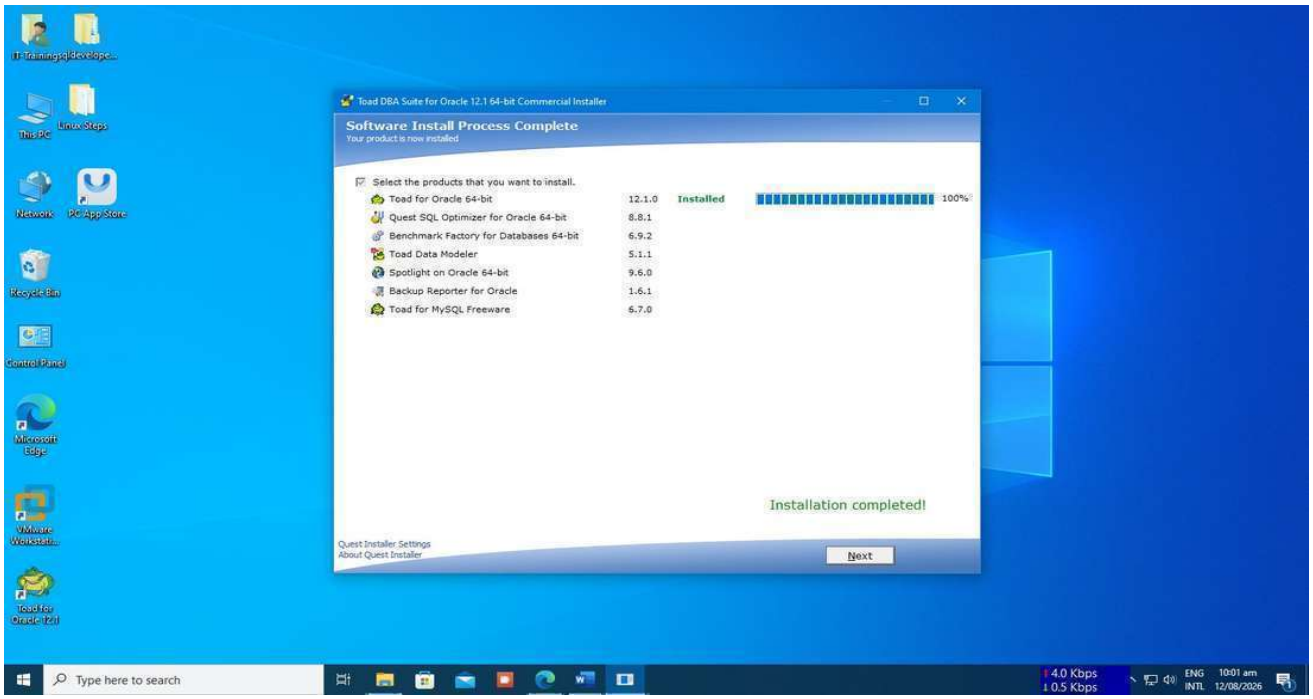
Review the installation settings.

Click:

Install

The installation process will begin.

Figure 6.8.6: Toad Installation in Progress



6.9 Entering License Key and Site Message

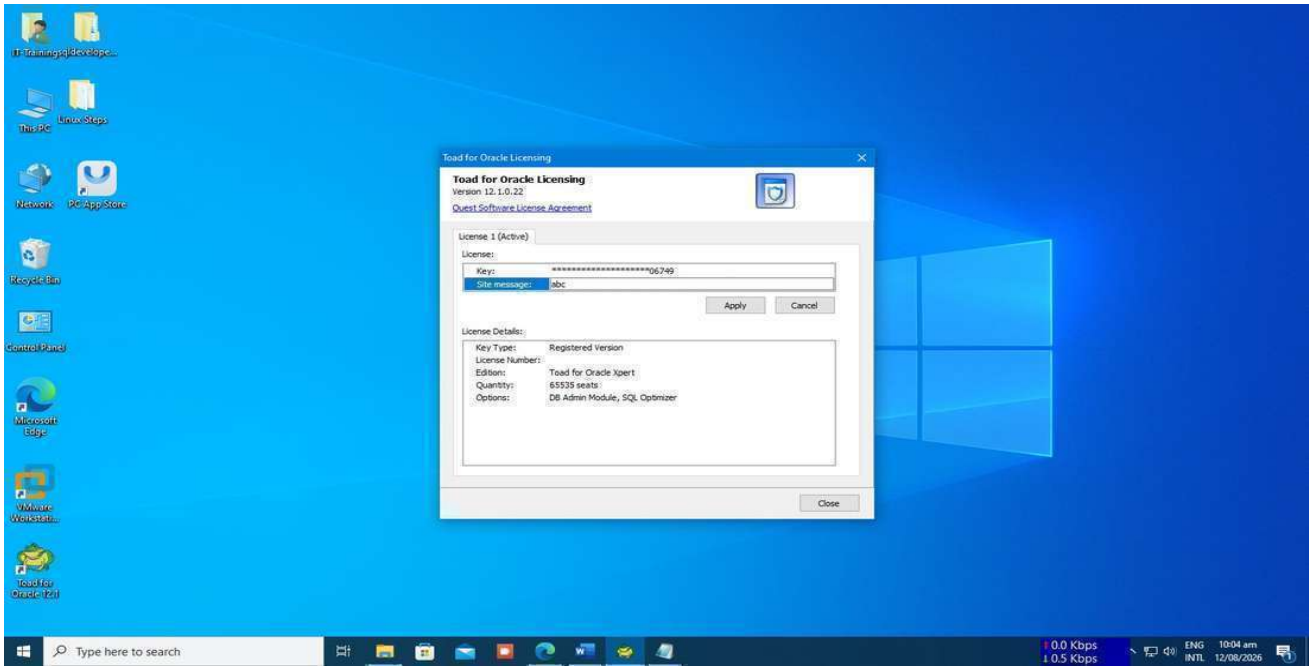
Since your instructor provided a **license key and Site Message**, we will document the activation step separately.

Step 1: Start Toad

Open :

Start Menu → Toad for Oracle

Figure 6.9.1: Toad Application



Step 2: License/Activation Window

If Toad displays a licensing or activation window, enter the information provided by the instructor.

License Key

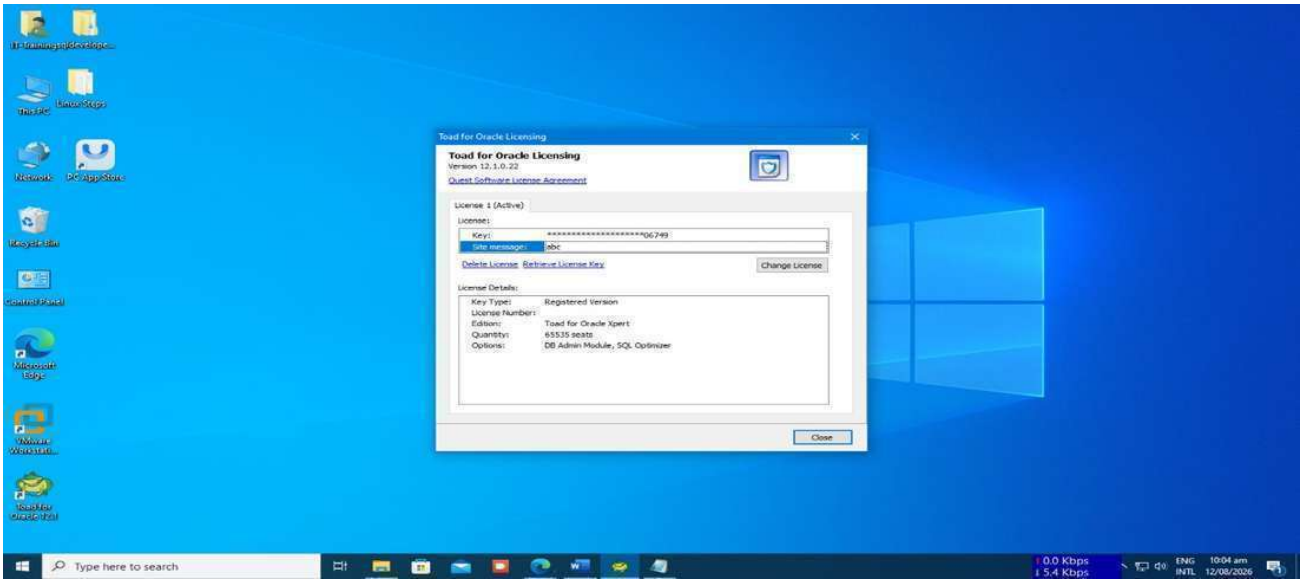
Enter the provided : 0-63920-01873-29060-06749

Site Message

Enter the provided : abc

Important: Do **not** put the actual license key or Site Message in the university report. Use a screenshot with the sensitive/license value hidden or blurred.

Figure 6.9.2: Toad License Activation



6.10 Launching Toad for Oracle

After successful installation and activation, launch Toad from the Windows Start Menu or desktop shortcut.

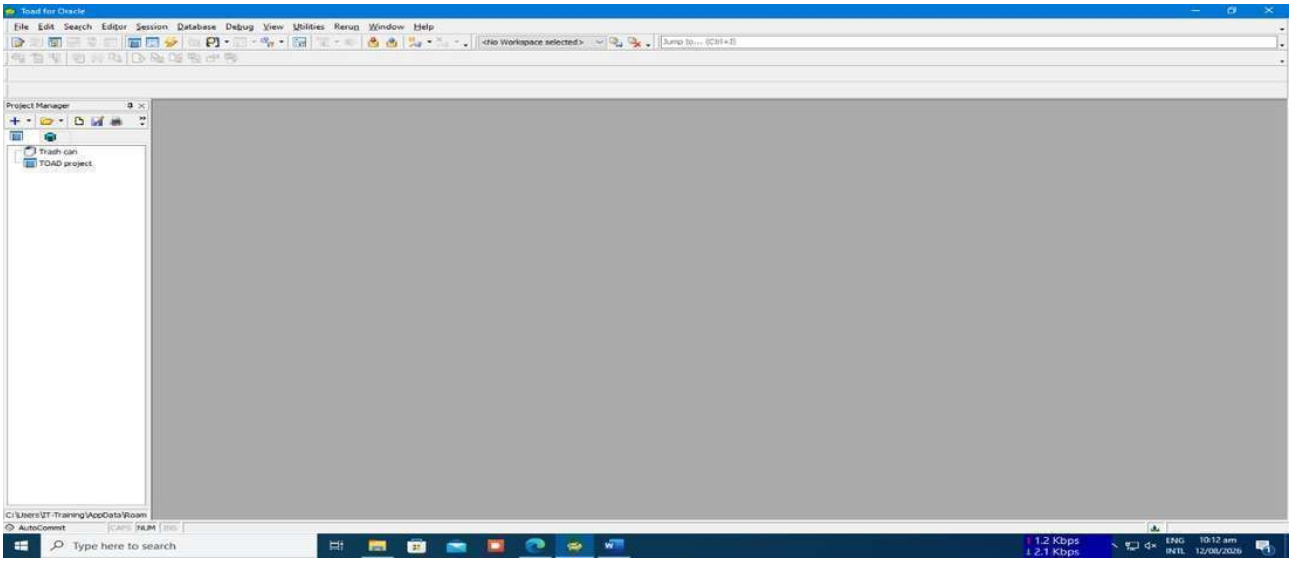
Command/Action

Start → Toad for Oracle

Expected Result

The main Toad interface will open.

Figure 6.10 : Toad Main Interface



6.11 Configuring Oracle Database Connection

Now Toad needs to be connected to the Oracle Database 12c running in the Oracle Linux virtual machine.

For our environment, the connection requires:

User : Oracle database user (system)
Password : Database password (oracle)
Host : Oracle Linux VM IP address (192.168.3.128)
Port : 1521
Service : pdborcl.virtual.pk

Example

Based on our Oracle environment, the connection information may look similar to:

Host : 192.168.x.x
Port : 1521
Service : orcl.virtual.pk

Important: Use the **actual IP address and service name shown by your current Oracle VM/listener**, rather than blindly copying the example above.

6.12 Connecting Toad with Oracle Database 12c

Step 1: Open Login Window

When Toad starts, open the Oracle login/connection window.

Enter the database credentials.

Example:

User: BOUTIQUE

Password: ********

or use another database user that your instructor assigned.

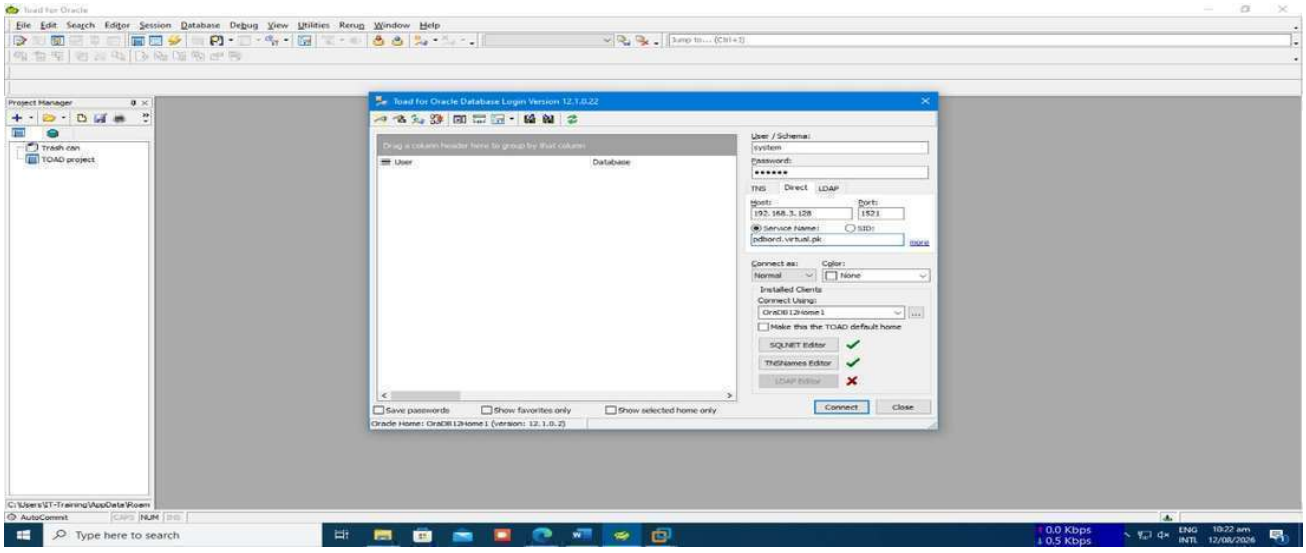
Step 2: Select Oracle Connection

Select the appropriate Oracle connection method/connection entry.

Enter:

Hostname : 192.168.3.128
Port : 1521
Service Name : pdborcl.virtual.pk
Username : system
Password : oracle

Figure 6.12.1 & 2Toad Oracle Connection Settings



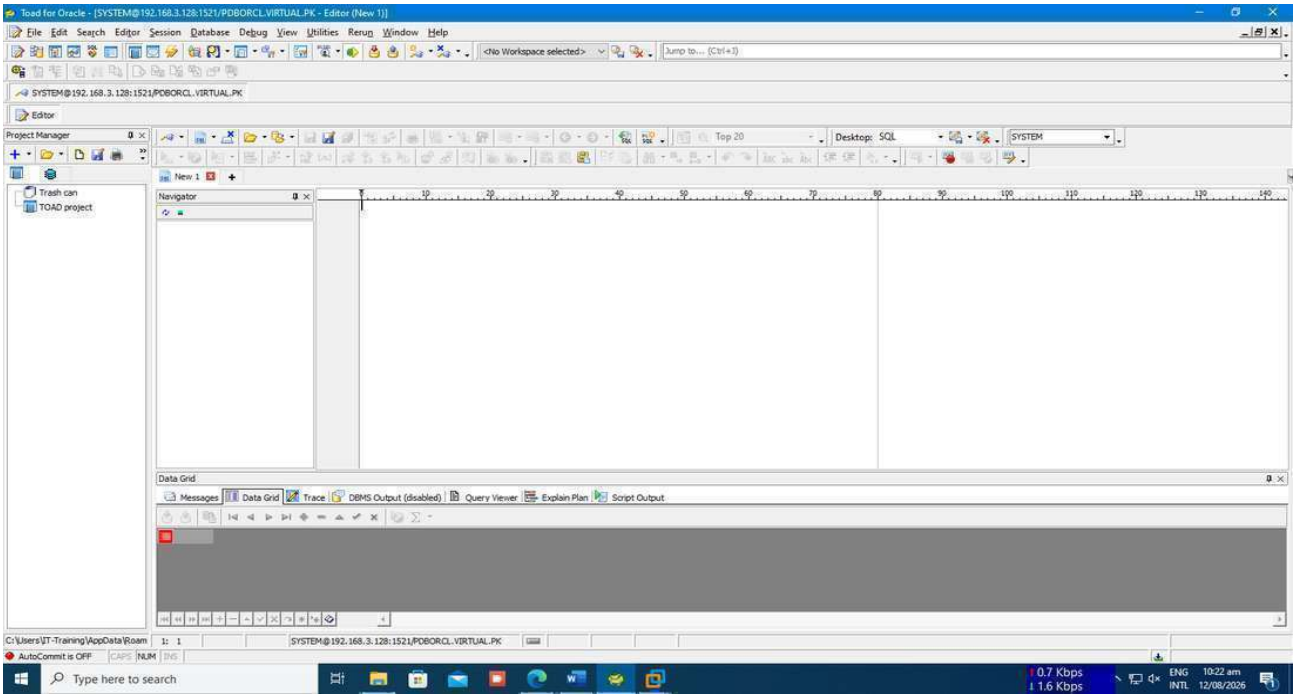
Step 3: Connect

After successful testing, click:

Connect

Toad will connect to the Oracle Database 12c instance.

Figure 6.13: Toad Connected to Oracle Database



***** Best of Luck *****